MUHAMMAD IMRAN

ZOO502-Animal-Physiology-and-Behavior

Solved Questions FINAL TERM (PAST PAPERS)

1. What is blubber and its role 2 marks

Blubber is a thick layer of fat, also called adipose tissue, directly under the skin of all marine mammals.
Role: As fatty tissues are little metabolically active & have little blood vascularization so it is a good insulator b/c of low thermal conductivity and prevent heat loss through body surface.

Marine mammals (seals, walruses, sirenians, cetaceans—whales) have thick layers of insulating subcutaneous fat which is known as blubber.
Blubber is a good insulator because it has a lower thermal conductivity.

Fatty tissues are metabolically very inactive and have little blood vascularization. This prevents heat loss through body surface.

2. Define Peristaltic 2 marks

Waves of contraction and relaxation in the smooth muscles present in the walls of alimentary canal. The alimentary canal starts from oral cavity and end at distal end of rectum. Esophagus conducts the swallowed bolus by peristaltic movement.
The esophagus conducts the bolus by peristaltic movement.
Peristaltic movements are the waves of contraction and relaxation in the smooth muscles of alimentary canal walls.
These movements s
tart from oral cavity and end at the distal end of rectum.

3. Write down the basic types of alimentary system 2.marks

Types of Alimentary Systems:
Two basic plans of alimentary systems in animals:

A. Sac-like gastro vascular cavity
B. Tubular alimentary canal or gut

4. How alveolar collapse is prevented? 2 marks

Prevention of Alveolar Collapse: Alveolar collapse normally does not occur for two reasons
1. Surrounding tissue prevents over-expansion of alveoli.
2. Presence of pulmonary surfactants.

5. Differentiate between laminar and turbulent flow? 2marks

Laminar Flow:
Flow is streamlined and continuous. The velocity of fluid is constant at every point. Flow occurs in layers at different velocities. Flow is zero at the wall and maximal at the center, A pressure difference supplies the force required to slide adjacent layers to past each other.
Turbulent Flow:
Direction of fluid movement is not aligned. It requires higher energy to move blood through Vessels. Turbulence also happens if smoothness is reduced by any obstruction e.g. Buildup of fatty deposits on arterial walls.

6. Name mechanoreceptor fiber in the arterial wall 2 marks

Atrial Receptors: Atrial walls have mechano-receptive afferent fibers including:

A. Myelinated A-type afferent fibers
B. Myelinated B-type afferent fibers
C. Unmyelinated C-type afferent fibers
D. Stretch-sensitive secretory cells

7. Fahraeus- lindqvist effect 2 marks

Fahraeus-Lindqvist Effect: Phenomenon of plasma skimming & resultant increased blood flow velocity is called Fahraeus Lindqvist effect. It reduces energy required to drive blood through microcirculation

8. Types of vascular resistance 2 marks

Vascular Resistance "The resistance that must be overcome to push blood through the circulatory system and create flow"
Types

1) Systemic vascular resistance (SVR)
2) Pulmonary vascular resistance (PVR)

9. Objectives of ethology 3 marks

Ethology:
"Scientific study of animal behavior in natural environment".
Objectives of Ethology: Involves understanding of

1. Understanding the stimuli that elicit behavior.

2. Understanding the physiological mechanisms that mediate the response.

3. Understanding how an animal's experience during growth and development influence the response.

4. Understanding how the behavior aids in survival and reproduction.

5. Understanding the behavior's evolutionary history.

10. Four accessory glands of elementary canal? 3 marks

Accessory Glands: Secrete digestive juices through ducts.
In mammals, the alimentary canal is associated with various accessory glands that secrete digestive juices through ducts into the gut.
The accessory glands of the mammalian digestive system are:

a. Three pairs of salivary glands
b. Gastric glands
c. Pancreas
d. Liver
e. Gallbladder

11. Briefly describe ornithine urea cycle 3 marks

Ornithine-Urea Cycle
• All vertebrates except most teleost fishes, synthesize urea primarily in the liver via the ornithine-urea cycle.
• In this cycle, two ammonia and one CO2 molecule are added to ornithine to form arginine.
• Arginine is cleaved by the enzyme arginase, to remove a molecule of urea and regenerate ornithine.

12. Factors affects systemic vascular resistance 3 marks

Factors affecting SVR
Elasticity of vessel wall
Diameter of vessel
Decreasing vessel diameter (vasoconstriction) increases SVR
Increasing vessel diameter (vasodilation) decreases SVR
Vascular resisance is determined mainly by the muscle tone in the smooth muscle tissue of the tunica media and the elasticity of the elastic fibers there.
Vasoconstriction (i.e., decrease in blood vessel diameter) increases SVR, whereas vasodilation (increase in diameter) decreases SVR.

13. 3 uses of crop in various animals. (3) marks

Crop
• A sac-like expanded section in esophagus of some animals. Used to store food without digesting.
Crop plays various roles in different animals. For example
• Food storage without digestion

• Fermenting the food
• Prepare food for nestlings (pigeon's milk)
In some animals crops are also used to ferment or digest foods for purposes other than normal digestive process. For example some parent birds prepare food in this way to be regurgitated for their nestlings e.g. pigeon's milk.

14. Name 3 vasodilators in the body? 3 marks

1. Histamine released from connective tissue and white blood cells in injured tissues causes vasodilation
2. Plasma kinins (e.g. bradykinin) are potent vasodilators in damaged tissues
3. Nitric oxide (NO) serves as a major inducer of vasodilation in cardiovascular system also causes relaxation of vascular smooth muscle and acts as vasodilator
4. Lactic acid generated by tissue cells
5. Rising conc. of K+ or H+ ions in interstitial fluid
6. Prostacyclin causes vasodilation and acts as an anticoagulant.

15. Write the name of local vasoconstrictor? 3 marks

1. Prosta gland and thromboxanes released by activated platelets and white blood cells.
2. Endotheline released by damaged endothelial cells.

If autoregulation fails to normalize conditions at the tissue level, neural mechanisms and endocrine factors are activated.

16. Write down the composition of the Bile? 3 marks

Bile: Vertebrate liver secretes bile. Bile doesn't contain any digestive enzymes It contains many salts Bile salts are essential for the digestion of fats

Composition of Bile:

Bile consists of water and a weakly basic mixture of cholesterol, lecithin, inorganic salts, organic salts, and bile pigments.

The bile organic salts are manufactured by the liver from cholesterol and amino acids complexed with sodium.

The bile pigments are derived from biliverdin and bilirubin, which are products of the breakdown of hemoglobin

coming from ruptured red blood cells.

17. What is Rete Mirabile? Mention its function in the Tuna fish? 5 marks

Rete Mirabile A highly evolved system of countercurrent heat exchange. Found widely in homeothermic and heterothermic vertebrates. The artery carrying warm blood toward extremity is completely encased in a circlet of veins carrying back cold blood.

Function in the Tuna fish:

Forms an extensive heat exchange surface b/w inflowing and outflowing blood. Large number of rete mirabile in tuna (fish) are used to regulate temperature of brain, muscles, and eyes. Swimbladders of some fishes (e.g. eels) are associated with rete mirabile and functioning as CO_2 exchangers.

A countercurrent network arrangement of arterial and venous capillaries is referred to as a rete mirabile. Before entering a tissue, an artery divides into a large number of small capillaries that parallel a series of venous capillaries leaving the tissue.
The arterial capillaries are surrounded by venous capillaries.

Rete Mirabile—Functions

This arrangement forms an extensive exchange surface between inflowing and outflowing blood.
Examples
Tuna (fish) have a large number of rete mirabile, which are used to regulate the temperature of the brain, muscles, and eyes.
The rete mirabile leading to the swimbladder of some fish (e.g. eels) function as a carbon dioxide countercurrent exchanger.

18. Note on physiological adaptation of ectothermic in hot environment 5 marks

Physiological Adaptations

- Ectotherms have high heat conductance
- Radiate heat rapidly—by moving to less warmer places
- Panting: Above 40oC many reptiles start panting—like birds and mammals
- Increases heat loss through respiratory evaporation, causing cooling.
- Regulating heart rate and flow of blood to surface tissues—to eliminate heat (e.g. marine iguanas)
- Gila monster (Heloderma suspectum) evaporates water from its cloaca to cool—a process similar to sweating in mammals

Most ectotherms have high heat conductance that allows them to radiate heat rapidly on moving to less warm places.

Above 40oC many reptiles start panting, just like birds and mammals to increase heat loss through respiratory evaporation that causes cooling.

The marine iguanas can regulate heart rate and flow of blood to the surface tissues to either absorb or eliminate heat.

The Gila monster (Heloderma suspectum) evaporates water from its cloaca to cool its internal body temperature—a process similar to sweating in mammals.

19. Write a note on types of stomach in vertebrates 5 marks

Types of Stomach in Vertebrates: Based on number of chambers, stomachs are classified as:

a. Monogastric stomachs
b. Digastric stomachs

Monogastric Stomachs

A monogastric stomach consists of a single strong muscular tube or sac.
Carnivorous and omnivorous vertebrates have a monogastric stomach.

Digastric Stomachs

Digastric stomachs are multichambered and are found in ruminant mammals of order Artiodactyla e.g. deer, giraffe,
sheep, cattle, camel, llama etc.
These stomachs have four chambers which are separated into two divisions.
• The first division consists of fermentation chambers called rumen and reticulum.
• The second division comprises the omasum and abomasum (true stomach).

20. What is hypoxia? Responses of aquatic animals in it 5 marks

Hypoxia • "A condition in which the body or a region of the body is deprived of adequate oxygen supply" **Hypoxia in Aquatic Environment**
• O2 levels in aquatic environment vary more frequently
• Aquatic animals are faced with frequent and rapid changes in oxygen levels. Many aquatic animals can withstand very long periods of hypoxia

Respiratory Responses of Aquatic Animals to Hypoxia

Oxygen level in aquatic environment varies frequently. Aquatic animals are subjected to more frequent and rapid changes in oxygen levels.

Many aquatic animals can withstand very long periods of hypoxia. Many of these animals utilize a variety of anaerobic metabolic pathways to survive the period of reduced oxygen availability.

Many animals adjust the respiratory and cardiovascular systems to maintain oxygen delivery during reduced oxygen availability. For example, many fishes increase gill ventilation rate in hypoxic conditions. Ram ventilators (e.g. tuna) swim forward with their mouth open, increase the size of gap during hypoxia to increase water flow over the gills.

21. What are pheromones? Also describe their Role? 10 marks

Pheromones:

• Chemical messengers released by many animals in environment to communicate. Spread through air.

Characteristics:

a. Affect the behavior of individuals of a species
b. Effective at remarkably low concentrations
c. Have particular odors
d. Detected by olfactory receptors of receiving animal

Roles of Pheromones:

1. Recognize Members of Species:
Unicellular organisms recognize members of their own species. Many insects identify members of their colony.

2. Reproductive Roles:
Pheromones attract members of opposite sex for breeding.
Induce courtship behaviors in members of a species.
Induce spawning behavior in aquatic animals with external fertilization.

3. Territorial Behaviors:
Many male mammals mark their territories with pheromones.
It warns other males of their occupied area.

4. Insect's Social Order:
In a honeybee colony, pheromones produced by queen maintain hive's complex social order.

5. Alarm Calls:
Members of some species produce pheromones on getting injured.
It warns other members by the presence of danger in area.

6. Repel Predators:

Pheromones produce by the pray may repel predators.
Example: The skunks produce foul-smelling musk that makes skunks unpalatable or distasteful to their enemies.

22.Define vestibular apparatus ? 2marks

Vertebrate Organ of Equilibrium

• Called as vestibular apparatus, located in the inner ear. It has:

1. • Saccule
2. • Utricle
3. • Semicircular canals

23.Abnormalities due to GH? 5 marks

Disturbances in the secretion of GH lead to abnormal growth and development in humans:

Gigantism

• Hypersecretion of GH before puberty causes excessive size and stature.

Developmental Abnormalities Due to GH

Acromegaly

• Hyper secretion of GH after maturity causes enlargement of bones of the head and limbs.

Dwarfism

• Insufficient secretion of GH during childhood and adolescence causes underdevelopment of The body (short stature).

24.Oxytocin? 3 marks

Is a type of harmone .

oxytocin are released by axon terminals in the posterior pituitary. Stimulates milk ejection, uterine contractions during child birth.

Neurohypophyseal hormones are synthesized and packaged in the cell bodies of two groups of neurosecretory cells in the anterior portion of the hypothalamus:

1. Supraoptic nuclei
2. Paraventricular nuclei

25. Five types of hormone ?

Based on their structure and pathway for synthesis, hormones are divided into four groups:

• Peptide and protein hormones
• Amine hormones
• Steroid hormones
• Prostaglandins

26.Gastrointestinal track? 5

Gastrointestinal tract

Cholecystokinin Stimulates secretion of enzymes by pancreatic acinar cells
Stimulates gall bladder contraction
Chymodenin Stimulates secretion of chymotrypsinogen from pancreas
Gastrin Stimulates HCl secretion in stomach
Secretin Stimulates secretion of bicarbonate
Substance P Acts as enteric neurotransmitter for feeling of pain
Motilin Increases motility of intestinal villi

27. Prostaglandins.10 marks

Prostaglandins are about 16 hormone-like local regulators substances that constitute a family of

cyclic, long-chain, unsaturated, hydroxy fatty acids.

Synthesis

Prostaglandins are produced by all or nearly all tissues. They are synthesized in membranes from arachidonic acid, which is produced by cleavage of membrane phospholipids by phospholipases.

Mode of Action

In some cases, they act locally as paracrine agents and in other cases they act on distant target

tissues in endocrine fashion. They have a rapid, short-lasting effect, similar to that of lipid-insoluble hormones.

They bind to cell-surface receptors linked to the cAMP pathway.

Functions

Prostaglandins have diverse actions on a variety of tissues, particularly involving smooth muscles.

1. Role in Fertilization

Prostaglandins present in semen aid in fertilization by reacting with the female cervical mucus to make it more receptive to sperm movement. They also stimulate the smooth muscles of the female's uterine wall to contract, helping sperm reach an egg.

2. Aid During Labor

Certain prostaglandins are secreted by the placenta at the onset of childbirth. They make the muscles of uterus more excitable, enhancing uterine contractions during labor.

3. Role in Immune Response

The damaged tissues, that induce immune response, produce prostaglandins. Some prostaglandins act as local regulators of inflammation, promoting fever and inflammation and also intensify the sensation of pain. They worsen pain by increasing nociceptor sensitivity to noxious stimuli. The interleukin-1, produced in immune response, causes fever by inducing the formation of a prostaglandin E2 which acts in the hypothalamus to

elicit the fever reaction. The anti-pyretic, anti-inflammatory and pain-relieving effects of aspirin and ibuprofen are actually due to the inhibition of prostaglandin synthesis by these drugs.

4. Blood Clot Formation

Prostaglandins also help the platelets to aggregate and form blood clots. This is the basis of use of aspirin by patients at risk for a heart attack due to formation of clots.

5. Stomach Lining

Prostaglandins also help maintain a protective lining in the stomach.As aspirin interferes with prostaglandin synthesis, long-term aspirin therapy can result in damage to this protective lining, causing debilitating stomach irritation.

6. Production of Erythropoietin

Several prostaglandins stimulate the production of erythropoietin by kidneys. Erythropoietin is a hormone that stimulates the production of erythrocytes.

28.neuromodulation?

A physiological process by which neurotransmitters released by a small group of neurons diffuse through large areas of the CNS and interact and modulate the effect of many neurons simultaneously.

- This is in contrast to the normal synaptic transmission. Neurotransmitters involved in altering the cellular properties and efficacy of synaptic transmission of multiple postsynaptic neurons.
- They act through Gprotein linked receptors that use voltage-gated ion channels for ion movements.
- Generation of response is slow, but lasts longer.

29.role of corpus leutum in pregnancy

If the egg is fertilized and implants, active corpus luteum is maintained.

- So progesterone and estrogen secretion continues until placenta fully develops to take over the production of these hormones.

• At this stage, corpus luteum degenerates. In many mammals, (e.g. rat) corpus luteum is stimulated by prolactin and continues to grow and secrete estrogen and Progesterone throughout the gestation period.

30.Types of stimuli

Two types:

• External stimuli Stimuli of odor, touch, light, sound and gravitation
• Internal stimuli Stimuli of pain, homeostatic imbalances and blood pressure

31.taste buds and taste receptors

The gustatory organs ofvertebrates are called taste buds.

A taste bud is composed of about 50 types of modified epithelial cells, including:

• Supporting cells (sustentacular cells)
• Basal cells
• Taste receptor cells
• basal cells are progenitor cells that give rise to new
• They regularly generate new sensory taste receptor cells which have an active life of only 10 days.

32.Feedback Mechanism (2)

Mechanism of self-regulation of Biological processes. Output or product itself regulates the process Maintenance of homeostasis

33.Acclimatization with example (2)

Acclimatization is a physiological, biochemical, or anatomic change within an individual animal that results from the animal's chronic exposure to new, naturally occurring environmental conditions.
Example.
Let's consider an animal that voluntarily migrates from a valley to a high mountain i.e. a voluntary change happens in its natural environment in which oxygen partial pressure is low.

34.Adrenal cortex hormone (3)

Adrenal (Cortex)
Aldosterone Increases blood Na+levels; increase K+secretion
Cortisol, corticosterone, cortisone
Role in carbohydrate metabolism increase blood glucose levels anti-inflammatory effects

35.Parts of eye take part in focusing (3)

The parts of the eye involved in focusing and image formation are:

• Cornea, a biconvex lens, pupil and retina.

36.Role of T3 and T4.10 marks

Metabolic Roles of T3 & T4

• They regulate bioenergetics by stimulating cellular respiration, oxygen consumption and metabolic rate.

• The acceleration of metabolism leads to a rise in heat production. This is of major importance in the thermoregulation of many vertebrates.

• They also sensitize some tissues to epinephrine that helps to maintain normal blood pressure, heart rate and muscle tone.

• They also promote normal motility of gastrointestinal tract and regulate reproductive

functions.

37.Acclimatization2 marks

An adaptive change within an individual animal.

• Happens due to chronic exposure to new, naturally occurring environmental conditions.

38.Rhodopsin 2 marks

The visual pigment in the rods is rhodopsin. Rhodopsins absorb photons of light energy and produce a generator potential.

Opsins are coupled to light-absorbing photopigment Retinal.

• This form a functional pigment molecule rhodopsin.

39.define line coding 2 marks

Each receptor subtype for taste sensations is connected to a particular set of axons in the nerve.

• In such an arrangement, information about one taste e.g. "sweetness" would be carried by some specific subset of axons.
• Such a pattern is called labeled line coding.

40. Explain the types of hair cells

Hair cells are ciliary cells.

• Found in several sensory organs of vertebrates.
• They are highly sensitive mechanoreceptors.
• Responsible for transducing mechanical stimuli into electrical signals.

Systems Based on Hair Cells: Examples

• Lateral-line system of fishes and amphibians, involved in detection of motion in the surrounding water.
• This system is based on hair cells.

Hair Cell Cilia

• Many cilia project from the apical end of each cell (reason of naming).
• Cilia are of two types:
• Kinocilium • Stereocilia
• A hair cell has a single kinocilium.
• Kinocilium has a "9 + 2" arrangement of internal microtubules (similar to other motile cilia).
• Each hair cell has 20-300nonmotile stereocilia.
• Stereocilia are structurally and developmentally distinct from the kinocilium.
• They are not formed of microtubules.
• They are formed of actin filaments.
• The stereocilia are arranged in order of increasing length from one side of the cell to the other.

41.diseases by hypothyroidism

Characterized by low production of throxine.

• It results from the lack of dietary iodine.
• It is represented by two types of diseases:
• Cretinism
• Goiter
• Cretinism results from iodine deficiency during early stages of development. In cretinism somatic, neural and sexual development is severely retarded, metabolic rate and resistance to infection is also reduced.
• Inadequate production of thyroxine in adults leads to goiter. In goiter, TSH is produced excessively that causes overstimulation of thyroid gland resulting in its enlargement
(hypertrophy).

42.Name glandular cells of thyriod and name their hormones too. (3marks)

Glandular Cells and Hormones of Adenohypophysis

1. Somatotropes—produce growth hormone (GH)
2. Corticotropes—produce adrenocorticotropic hormone (ACTH)
3. Thyrotropes—produce thyroid-stimulating hormone (TSH)
4. Gonadotropes—produce gonadotropic hormones (luteinizing hormone, LH and follicle stimulating hormone, FSH)
5. Lactotropes—produce prolactin (PRL)

43.Range fractionation

In a sense organ, sensory receptors are arranged in an order of increasing sensitivity to different range of intensities of the stimulus.

- This hierarchical arrangement of receptors is known as range fractionation. Each individual receptor, in a sense organ, covers only a fraction of the total dynamic range of the stimulus intensities.
- Receptors work together in a hierarchical way to provide discrimination of stimulus intensities.

44.Placental hormone

In pregnancy, placenta secretes large quantities of hormones, essential for maintenance of pregnancy.

- These hormones include:
- Human chorionic gonadotropin
- Estrogens
- Progesterone
- Human chorionic somatomammotropin

45.Progesteron

Progesterone is secreted in tremendous quantities by placenta.

- This secretion is about 10-fold more than normal levels. Causes development of decidual cells in endometrium. These cells play an important role in the nutrition of the early embryo.
- It decreases contractility of the uterus to prevent spontaneous abortion.

46.Color blindness 3

Color blindness is caused due to a mutation in one of the cone opsin genes that results in the absence of one type of pigmented cones.

- A person missing a single type of color receptive cones is unable to distinguish some colors.

47.Three use of crop?

A sac-like expanded section in esophagus of some animals. Used to store food without digesting. • Generally found in infrequent or quick feeders • Allows quantities of food to be stored for digestion at a later time

48.Define peristaltic?

The wavelike muscular contractions in tubular structures, especially organs of the digestive system such as the esophagus and the intestines. **Peristalsis** is characterized by alternate contraction and relaxation, which pushes ingested food through the digestive tract towards its release at the anus.

49.Objective of ethology?

Ethology is the scientific and **objective** study of animal behaviour, usually with a focus on behaviour under natural conditions, and viewing behaviour as an evolutionarily adaptive trait.

50.Three local vasodilator?

Vasodilation is the widening of blood vessels. It results from relaxation of smooth muscle cells within the vessel walls, in particular in the large veins, large arteries, and smaller arterioles. ...

- sildenafil (Viagra)
- tadalafil (Cialis)
- vardenafil (Levitra)

51.Three respiratory pigments?

A **respiratory pigment** is a molecule, such as hemoglobin in humans and other vertebrates, that increases the oxygen-carrying capacity of the blood. The four most common invertebrate **respiratory pigments** are hemoglobin, hemocyanin, hemerythrin and chlorocruorin.

52.Differentiate between Norepinephrine and Epinephrine?2

Norepinephrine is the primary excitatory transmitter in postganglionic cells of sympathetic system.

Epinephrine is excitatory at some synapses and inhibitory at others depending on postsynaptic membrane

53.Define vestibular apparatus?

The vertebrate organ of equilibrium is the vestibular apparatus. Vestibular apparatus, located in the inner ear. It has:

..The saccule

- The utricle

• Three semicircular canals lying orthogonally in three mutually perpendicular planes.

54.Name two components of rhodopsin ? 2mks

A rhodopsin molecule consists of two major components:

Opsin protein

A light-absorbing molecule (retinal or 3dehydroretinal) Function of LH in male and female.

In Females LH stimulates production of estrogen and progesterone by ovary.

It causes ovulation and formation of corpus luteum.

In males, LH is also called as ICSH (interstitial cell stimulating hormone) It stimulates testosterone production by the Leydig cells of testes.

55.What are gonadotropin

Gonadotropins: Luteinizing and Follicle Stimulating Hormones. Luteinizinghormone (LH) and follicle-stimulating hormone (FSH) are called gonadotropinsbecause stimulate the gonads - in males, the testes, and in females, the ovaries.

56.Define the terms gland and secretion? 2mks

A gland is a cell or group of cells that secretes a particular chemical substance for use in the body or for discharge into the surroundings.

Secretions are chemical substances, synthesized by glandular cells. They are released from the gland in response to an appropriate stimulus.

57.Write the role of Fibers of Zonula and ciliary muscles in eye? 3

Fibers of zonula held the lens in place. They change the shape of lens by exerting an outward tension on perimeter. Attached with fibers of zonula are the ciliary muscles
They contract to pull fibers of zonula, which flatten the lens. This focuses distant objects on retina. When they relax, lens becomes rounded and near object is focused.

58.Describe Hodgkin cycle in detail? 3

The number of Na+ channels open at any instant depends on Vm as well as on time i.e. the phase

of action potential. Thus changes in the conductance of sodium ions (gNa) occur as a function of Vm and time and reflect the behavior of thousands of Na+ channels, each one opening and closing during depolarization in accord with certain principles. During an AP, Na+ channels respond to an initial depolarization by opening, allowing Na+to enter the cell, which further depolarize the membrane. This depolarization causes more channels to open, allowing still more Na+to enter the cell and triggering an explosive, regenerative event. This relationship between membrane potential and sodium conductance is termed as the Hodgkin cycle and represents a type of positive-feedback system.

A stimulus causes local depolarization of the membrane. Na+ channels open in response to initial depolarization. This increases Na+ conductance that allows Na+ to enter the cell, which cause further depolarization of membrane and decrease in Vm. This relationship between Na+ conductance (gNa) and Vm represents a type of positive feedback system and is termed as the Hodgkin cycle.

It results in triggering of an action potential.

59.Which hormones are involved in the regulation of water and ions? 3

These hormones include:

- Antidiuretic hormone (ADH)
- Aldosterone
- Atrial natriuretic Hormone
- Calcitonin
- Parathormone

60.Messiner corpuscales and paccin corpuscales difference?3

Touch receptors in ridges of fingertips are Meissner's Corpuscles. They have encapsulated nerve endings. To receive pressure stimulus, Pacinian corpuscles are situated deep in skin, in the limbs they receive vibrations.

61.Basic components of homeostasis and example of homeostasis, Its mechanism (5max)

Examples of Homeostasis:

Homeostatic regulation of water • Homeostatic regulation of temperature • Regulation of pH • Regulation of glucose concentration • Regulation of osmotic pressure • Regulation of oxygen level • Regulation of ion conc.

Mechanism
Variables

Set point

Living control systems VS physical control systems

Three components: Receptors, Control centre, effectors

Example

• Temperature control system in air conditioners and water heating geysers VS Endothermic animals

62.Name glandular cells of thyriod and name their hormones too. (3mqx)

It contains five types of glandular cells that synthesize and secrete six hormones

1-Somatotropes produce growth hormone
(GH)

2-Corticotropes produce adrenocorticotropic hormone
(ACTH)

3-Thyrotropes
Produce thyroid-stimulating hormone (TSH)

4-Gonadotropes
Produce gonadotropic hormones (luteinizing hormone, LH and follicle stimulating hormone, FSH)

5-Lactotropes
Produce prolactin (PRL)

63.Exocirne gland with 2 examples atleast (5max)

Exocrine glands produce fluid secretions that are delivered through ducts onto the epithelial surfaces of the body. The fluid secretions may be proteins (enzymes) or mucous or both.

Examples:
Salivary glands produce saliva that is delivered to the oral cavity through submandibular and parotid ducts.

Pancreas produces enzyme-containing pancreatic juice that is delivered to the small intestine through pancreatic duct.

Lacrimal glands produce tears that is delivered through lacrimal duct on the surface of eye.

Mammary glands produce milk that is delivered through lactiferous ducts to the nipples

Androgens secreted by adrenal cortex 2 marks
Androgens secreted by adrenal cortex include:

1. Testosterone
2. Dihydrotestosterone
3. Androstenedione
4. Dehydroepiandrosterone (DHEA).

64.Effects of neuro-modulation 3 marks

- Development of complex behavioral patterns e.g. happiness, exploration, revenge, reward, greed.
- Processes of thinking, cognition, planning, learning and memory
- Behavioral problems: mood swings, sleep disturbances, feelings of stress, anxiety, anger and depression.

65. 2 examples of interoceptive receptors 3 marks

These include:

Receptors of vestibular system that monitor orientation of the body

Thermo-receptors that keep track of thermal state of the body

Chemo-receptors that keep track of chemical state of the body Proprio-ceptors that monitor position of the muscles and joints Nociceptors perceive the sensation of pain.
Pacinian corpuscles receive pressure stimulus.

66.Name and explain hair cell types 5 marks

Inner hair cells (IHC)

There are about 3,500 IHCs arranged in a straight line or wide U. They are true sensory cells which send impulses via auditory nerve.

Outer hair cells (OHC)

About 12,000-25,000 OHCs in mammals, they are arranged in three or four rows forming a characteristic W shape.

Outer hair cells have both sensory and motor elements. They contribute to hearing sensitivity and frequency selectivity

67.Hormones of ovary

LH stimulates production of estrogen and progesterone by ovary.

68.Explain photoreceptor

Photoreceptors can detect and generate response to a single photon of light. Photoreceptors possess light-sensitive carotenoid pigments retinal and 3dehydroretinal. Carotenoids are associated with opsin proteins to form rhodopsins. Rhodopsins absorb photons of light energy and produce a generator potential

69.Hair cilia types

Many cilia project from the apical end of each cell (reason of naming). • Cilia are of two types:

• Kinocilium • Stereocilia

A hair cell has a single kinocilium. • Kinocilium has a "9 + 2" arrangement of internal microtubules (similar to other motile cilia). • Each hair cell has 20-300nonmotile stereocilia. • Stereocilia are structurally and developmentally distinct from the kinocilium. • They are not formed of microtubules. • They are formed of actin filaments. • The stereocilia are arranged in order of increasing length from one side of the cell to the other.

70.Diseases by hypothyroidism

It is represented by two types of diseases: • Cretinism • Goiter

Cretinism results from iodine deficiency during early stages of development Inadequate production of thyroxine in adults leads to goiter.

71.active potential phases Stigma and ocelli 3

Eyespot or Stigma

Simplest photoreceptive structure. Found in some protozoa e.g. euglena. It is a bright red colored organelle and has carotenoid pigments. It gives a sense of light and dark. Helps in phototaxis.

Eyecups or Ocelli

Multi-cellular photoreceptive structure that consists of a cuplike depression containing photoreceptor cells; Found in cnidarians and flatworms e.g. Planaria.

Cannot form image, provides the animal a sense of direction only

72Functions of T3 and T4 and Hyperthyroidism 10. marks

These hormones have diverse effects on the physiology of virtually all tissues of the body. Their roles can be grouped into two categories:

1. Metabolic roles
2. Developmental roles

Metabolic Roles of T3 & T4

They stimulate cellular respiration, oxygen consumption and metabolic rate and, in turn, heat production. (important in thermoregulation).

They sensitize some tissues to epinephrine that helps to maintain normal blood pressure, heart rate and muscle tone.

They also promote normal motility of gastrointestinal tract.

They also regulate reproductive functions.

Developmental Roles of T3 & T4

They significantly affect the development and maturation of vertebrate animals.

They are involved in the normal functioning of bone-forming cells and the branching of nerve cells during embryonic development of the brain.

They control metamorphosis of a tadpole larva into adult frog.

Developmental effects of growth hormone occur only in the presence of thyroid hormones.

Hyperthyroidism

Excessive secretion of thyroid hormone is known as hyperthyroidism.

The most common form of hyperthyroidism is Graves' disease. It is an autoimmune disorder in

which antibodies that mimic TSH bind to the receptor for TSH and cause sustained thyroxine

production.

It leads to high body temperature, profuse sweating, weight loss, irritability, high blood pressure

and protruding eyes (exophthalmia).

BEST OF LUCK